

Climate Change Governance: Science, Data & Models

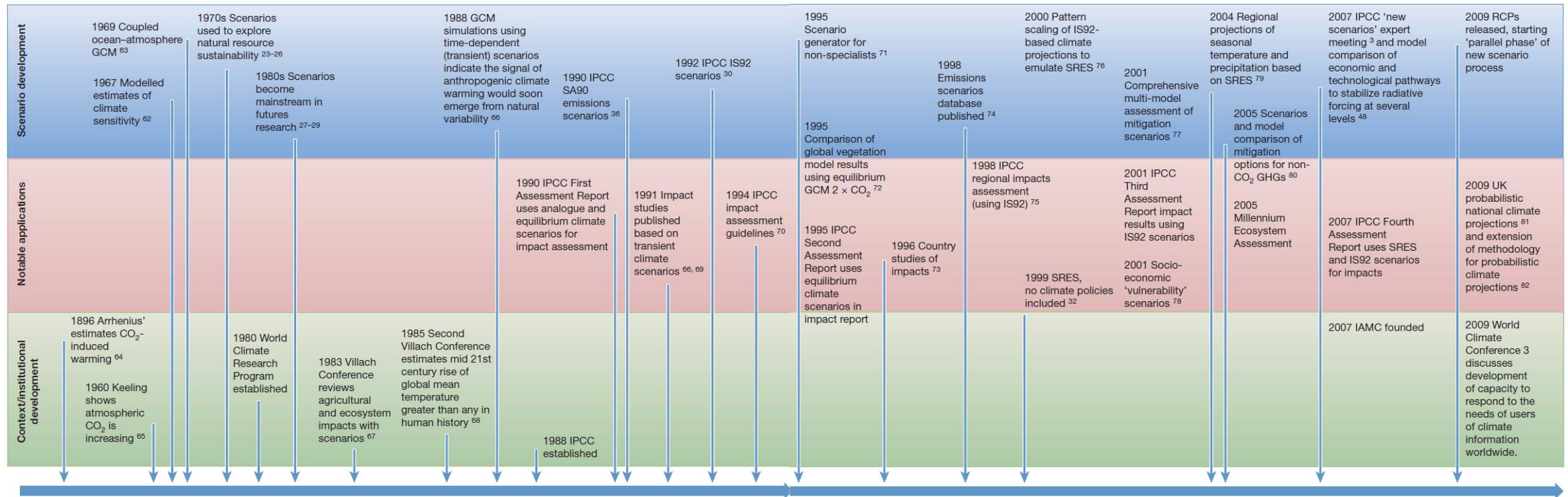


Lecture 7: RCPs and SSPs

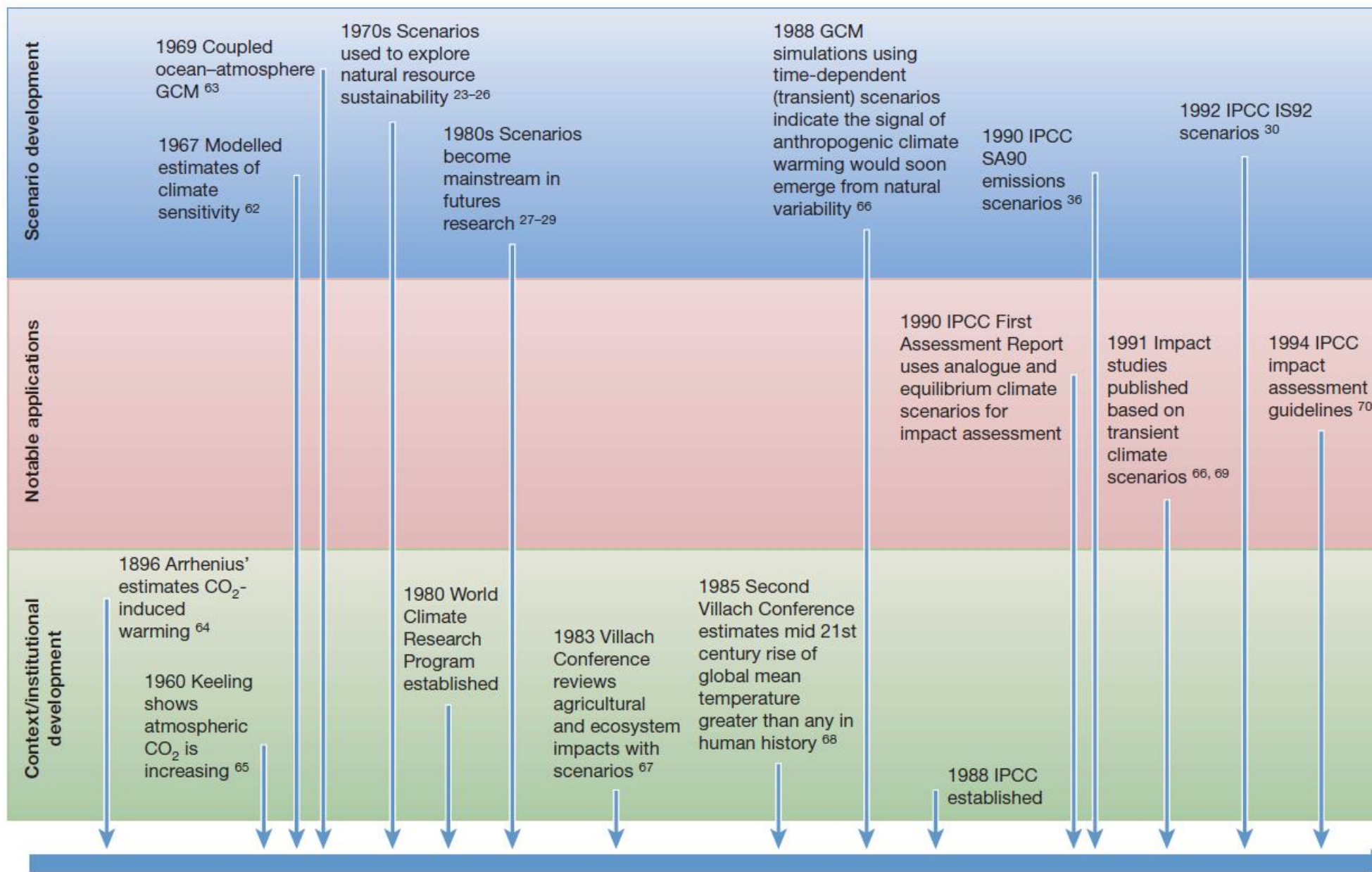
ENGG562/ENV562 – Fall 2026

Muhammad Awais

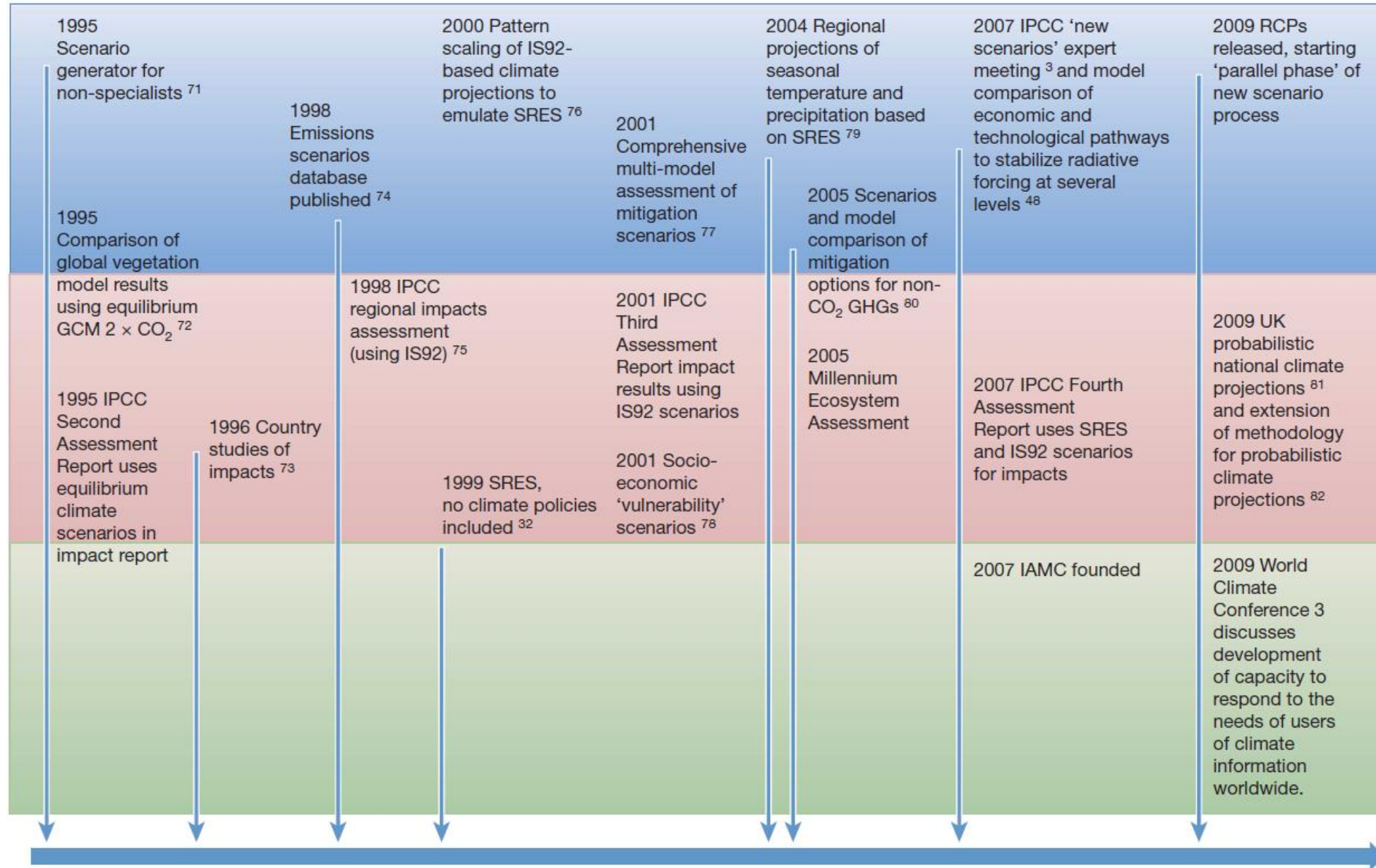
History of (IPCC) scenarios (1896-2009)



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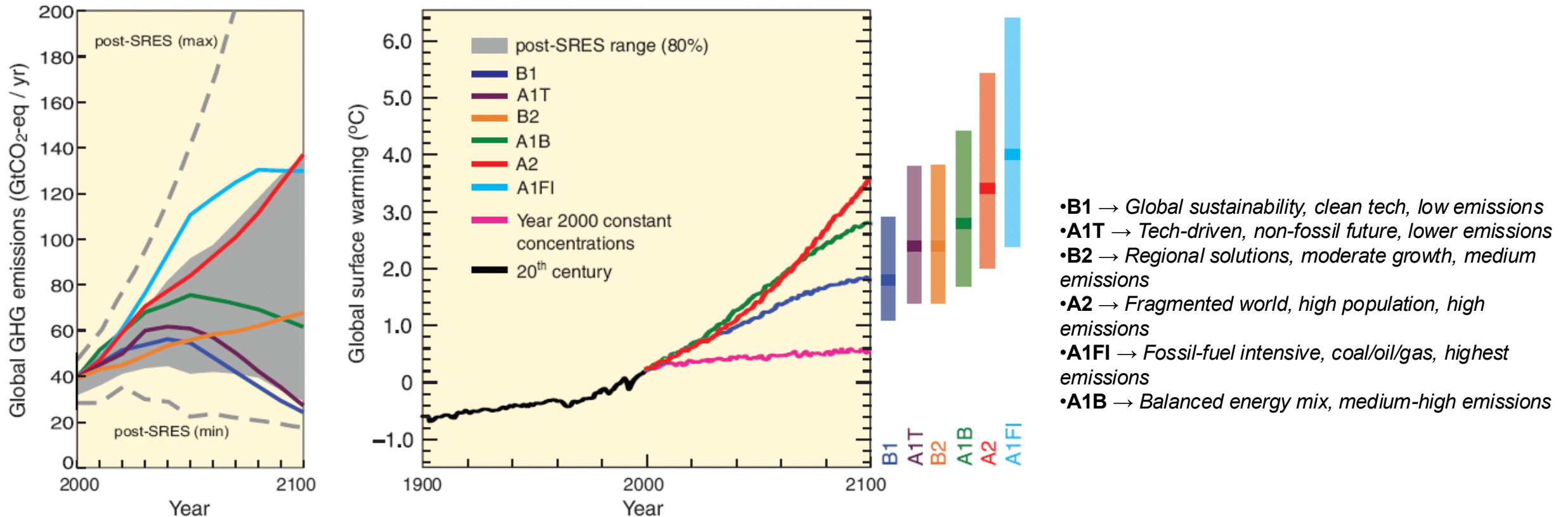
History of (IPCC) scenarios (1995-2009)



The “New Scenarios Process”

Old scenarios were qualitative in nature and more quantification were needed across different dimensions

Scenarios for GHG emissions from 2000 to 2100 (in the absence of additional climate policies)
and projections of surface temperatures

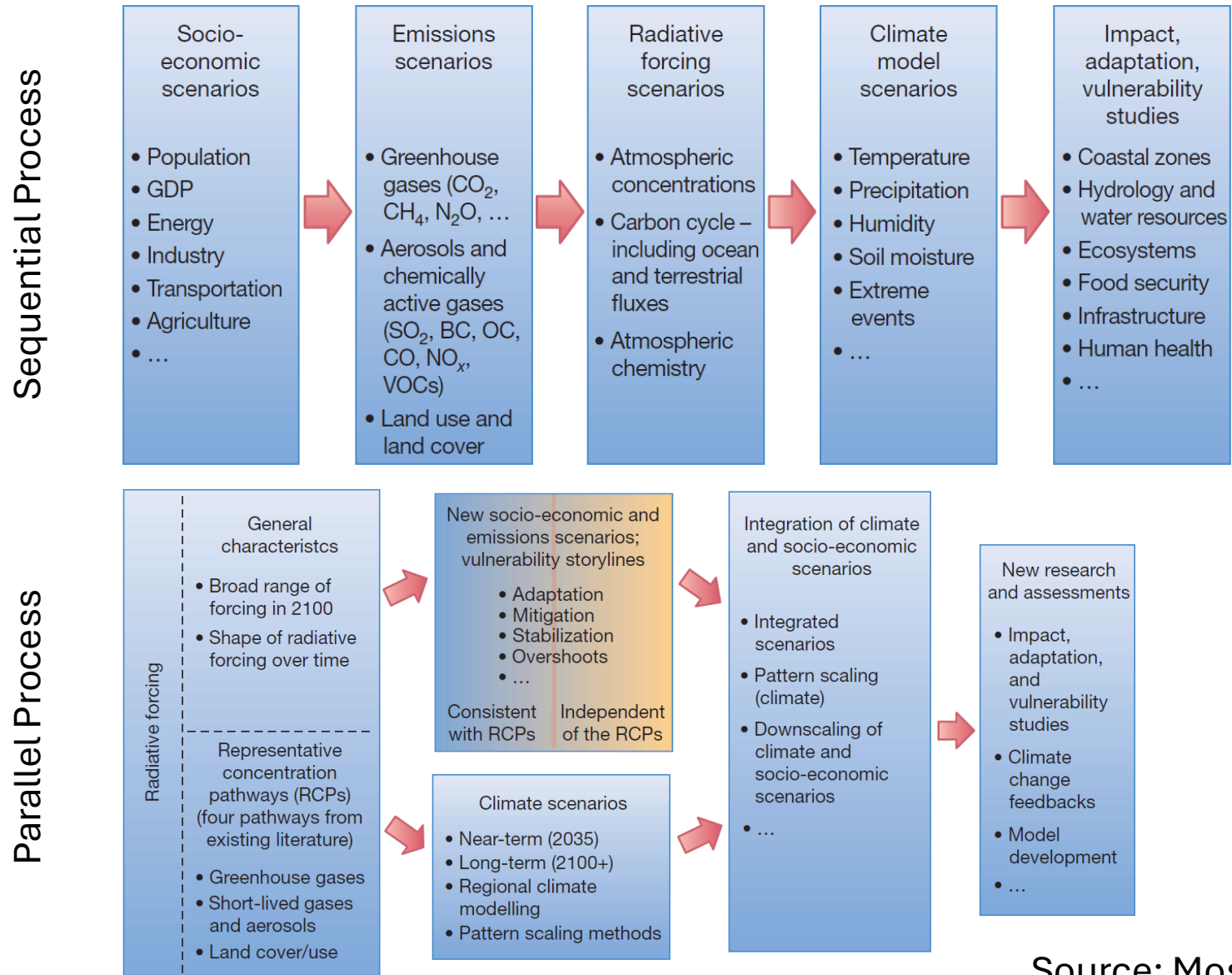


Four important reasons to develop new community scenarios for climate assessment:

- 1. Need to cover [a wider range](#) of GHG concentrations (SRES only included baseline scenarios)
- 2. Need for a [wider set of parameters](#) (Climate models have become more complex; higher information need).
- 3. Need for scenarios that cover mitigation & adaptation issues (need for [more collaboration](#) between “WGs”)
- 4. Use more recent insight into trends in scenario drivers ([update](#))

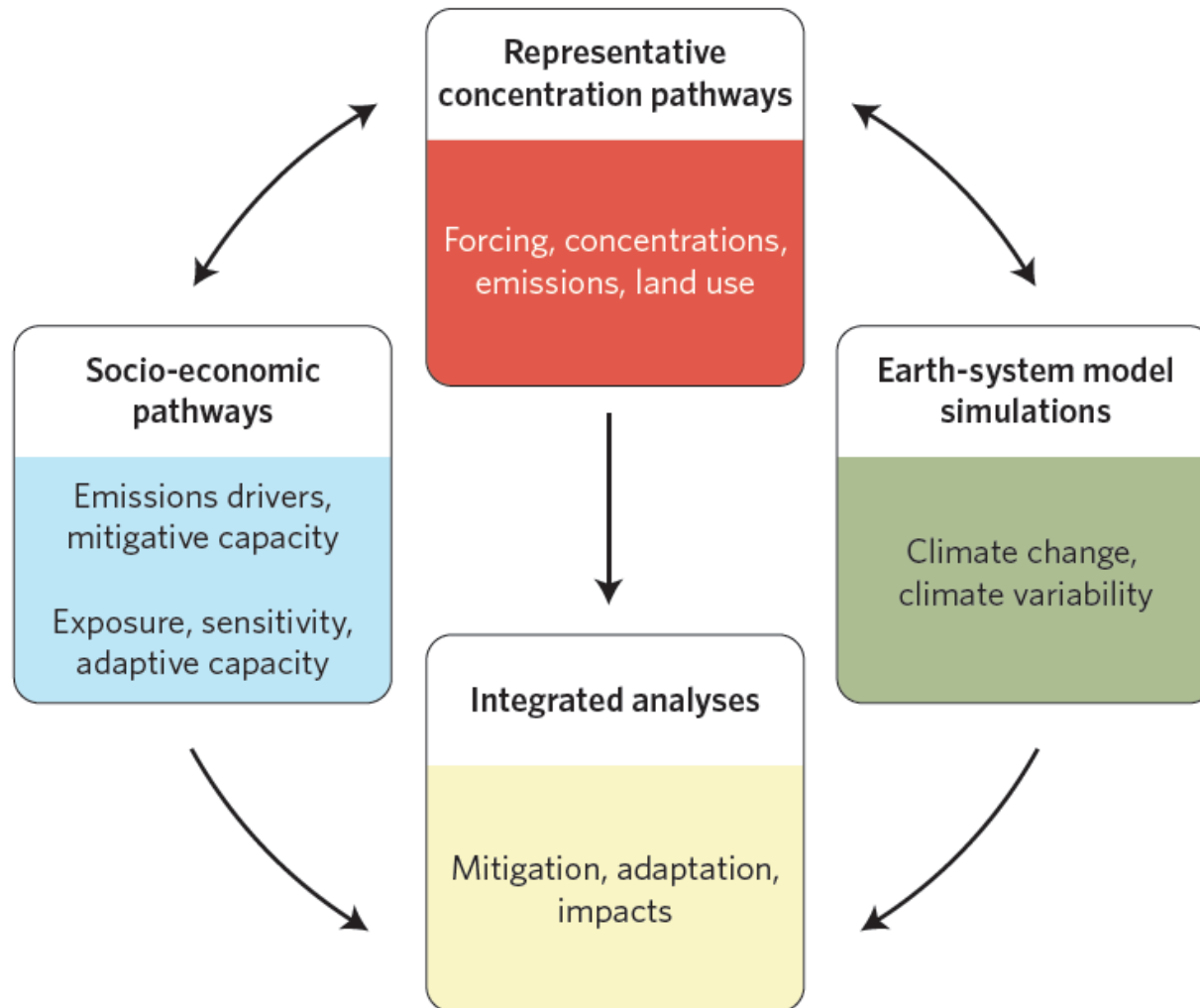
Moss et al. (2010) The next generation of scenarios for climate change research and assessment. Nature 463:747-756.

Sequential vs. Parallel Process



Source: Moss et al. 2010

The Parallel Process



Source: O'Neill & Schweizer, 2011

What are scenarios, why do we use them?

Scenarios.

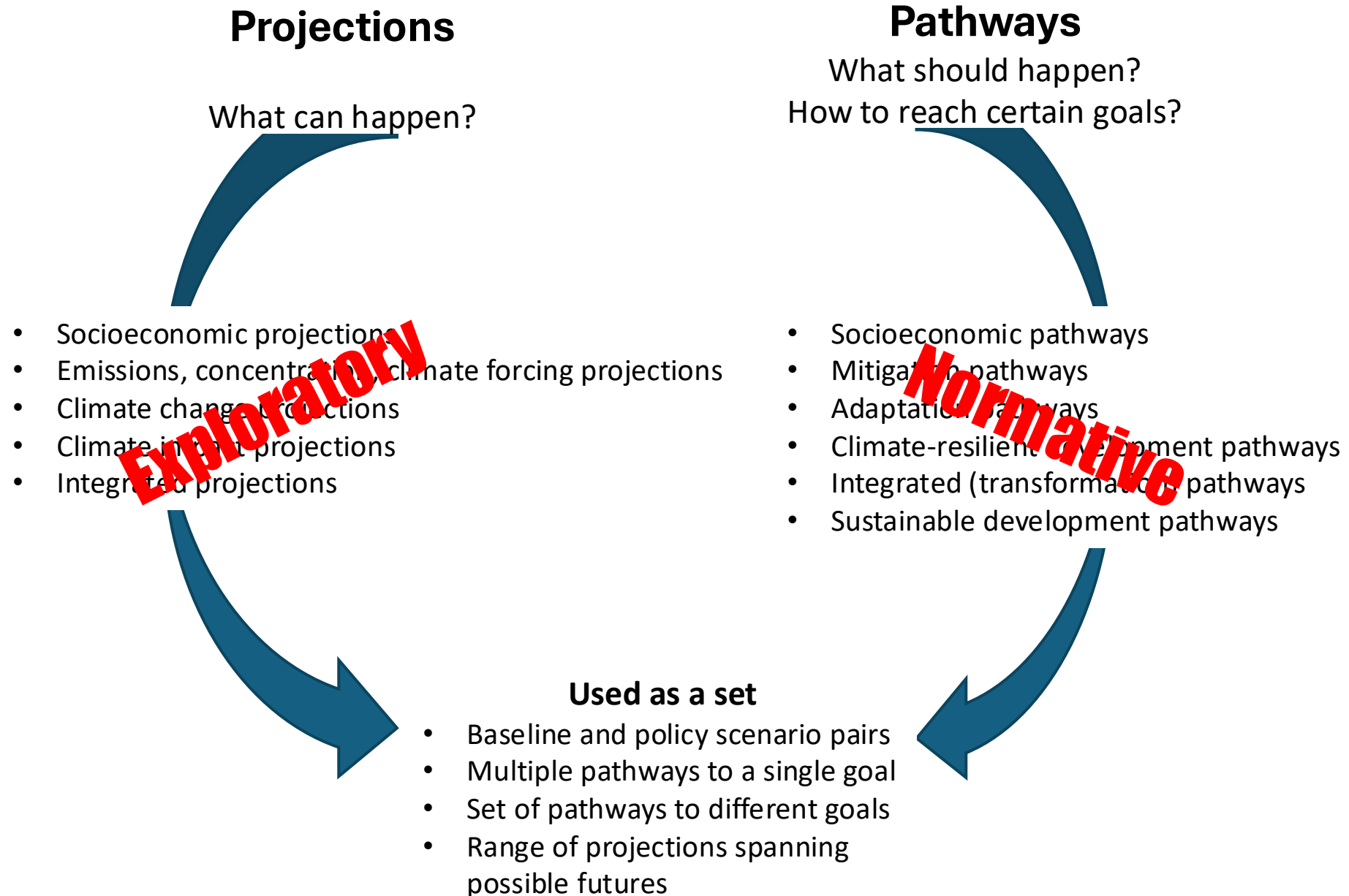
A scenario is a coherent, internally consistent, and **plausible** description of a possible future state of the world.

Scenarios are **not predictions or forecasts** (which indicate outcomes considered most likely) but are alternative images without ascribed likelihoods of how the future might unfold.

They may be **qualitative, quantitative, or both**.

An **overarching logic** often relates several components of a scenario, for example a storyline and/or projections of elements of a system.

IPCC: Climate Change Scenarios



Source: Elmar Kriegler, SENSES project

Scenarios are what-if storylines about future

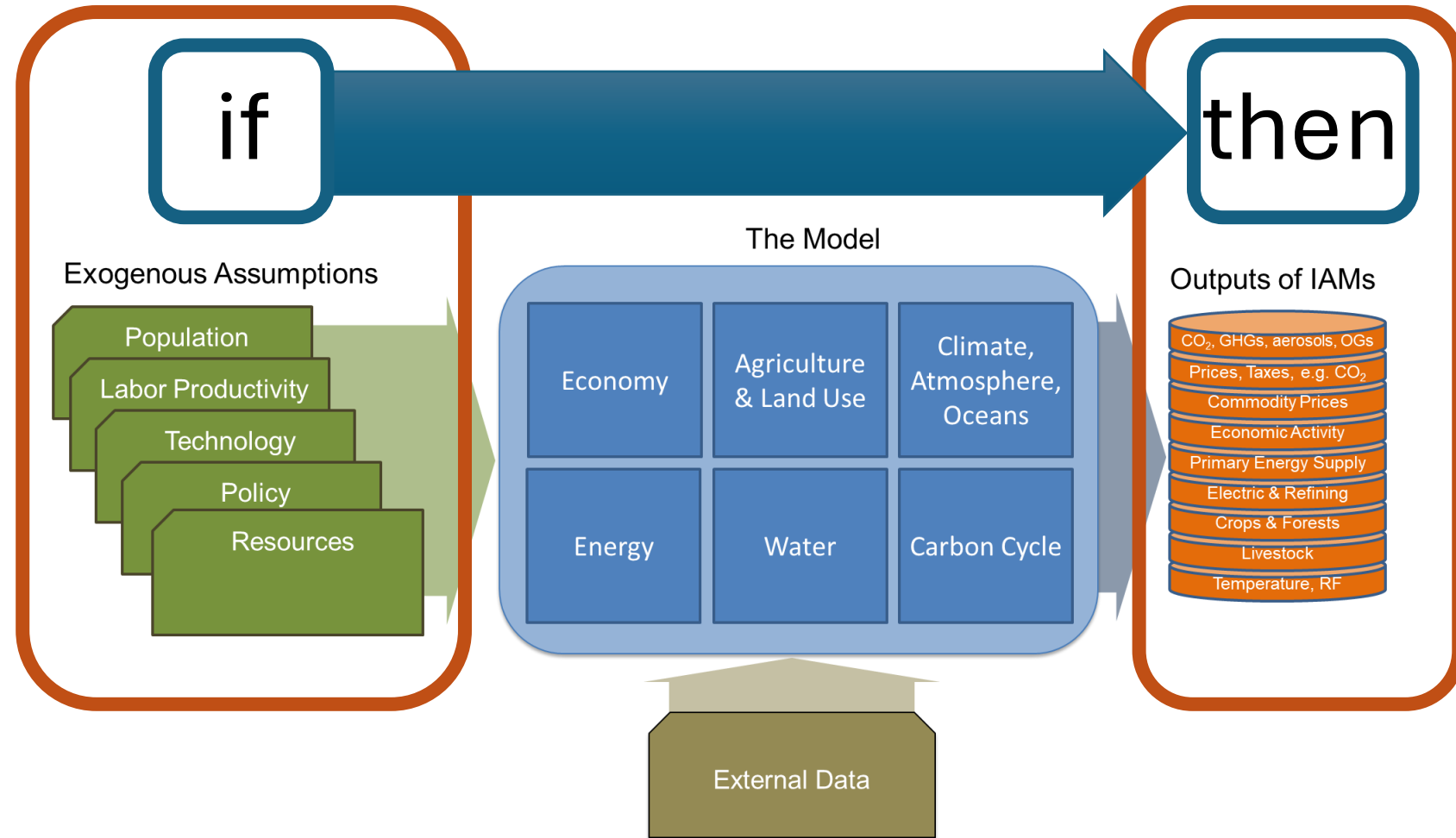
- They describe different possible futures under certain assumptions (e.g., population, economy, technology, or policies).
- They are not predictions, but rather a way to explore uncertainty of future.
- **Examples:**
 - *Everyday:* Imagine a scenario where **50%** of students at LUMS commute by public transport instead of private vehicles.
 - *In context of Climate Change:* Imagine a scenario where Pakistan decides that **50%** of all new motorbike sales must be electric.

Projections are model results that show what could happen *if a given scenario comes true*

- They are the numerical outcomes, such as changes in temperature, rainfall, or energy use.
- **Examples:**
 - *A simple example would be* If you save **PKR100** each day, the projection is that you will have **PKR3,000** at the end of the month.
 - In context of climate change, If average summer temperature in Lahore is **40°C**, a projection under a high fossil-fuel use scenario might show it rising to about **44–45°C** by the end of the century.

Models and Scenarios

- Models are structured relationships that produce **conditional forecasts**.
- Scenarios are conditional forecasts.



Representative Concentration Pathways (RCPs)

Two main types of models are used in Climate Science

Integrated Assessment Models (IAMs)

- Represent **human systems** (energy, land, economy, policy).
- Explore how choices (technology, policy, lifestyles) shape future **emissions**.
- Output: emissions pathways, energy and land-use projections etc.

Global Climate Models (GCMs)

- Represent the **Earth system** (atmosphere, oceans, land, ice).
- Use physics and chemistry to simulate the climate's response to emissions.
- Output: temperature, rainfall, sea level, extreme events etc.

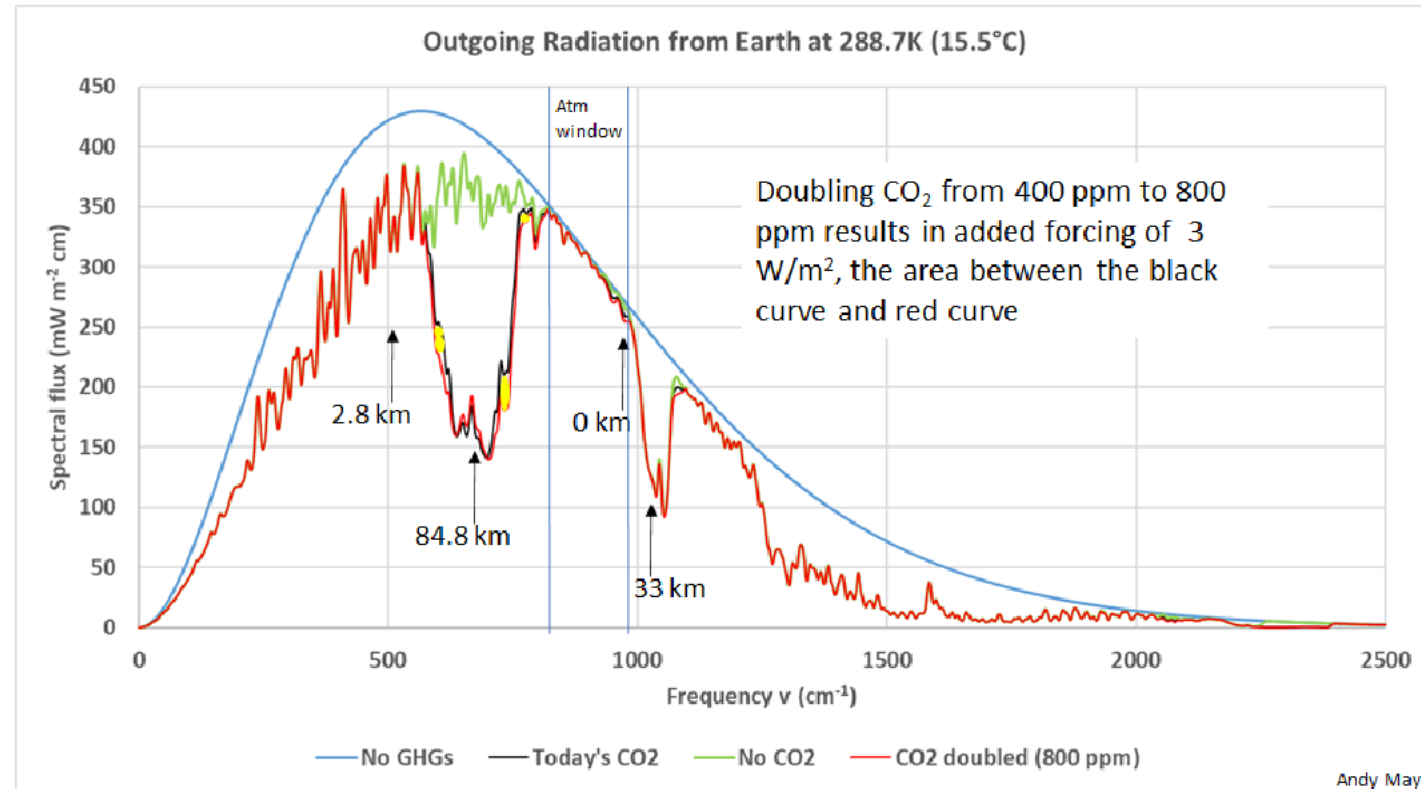
RCPs –Future greenhouse gas concentration trajectories used by climate models

- These are concentration pathways and not just emissions → defined by radiative forcing (extra energy trapped in earth's atmosphere)
- Higher forcing --> more trapped heat → higher global temperatures

Representative → not exact predictions, but *typical examples* of possible futures.

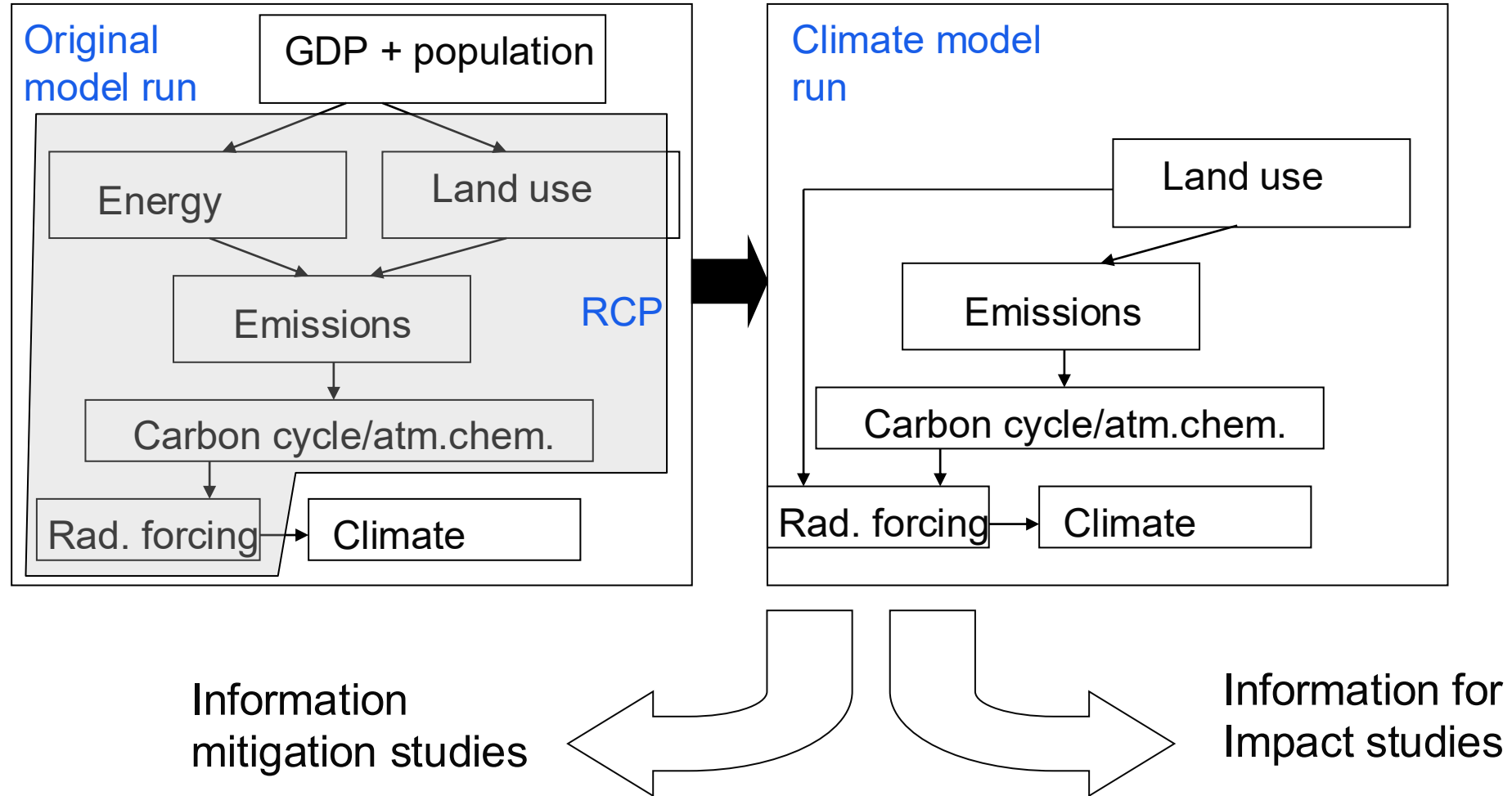
Concentration → describes the **greenhouse gas concentration levels** in the atmosphere.

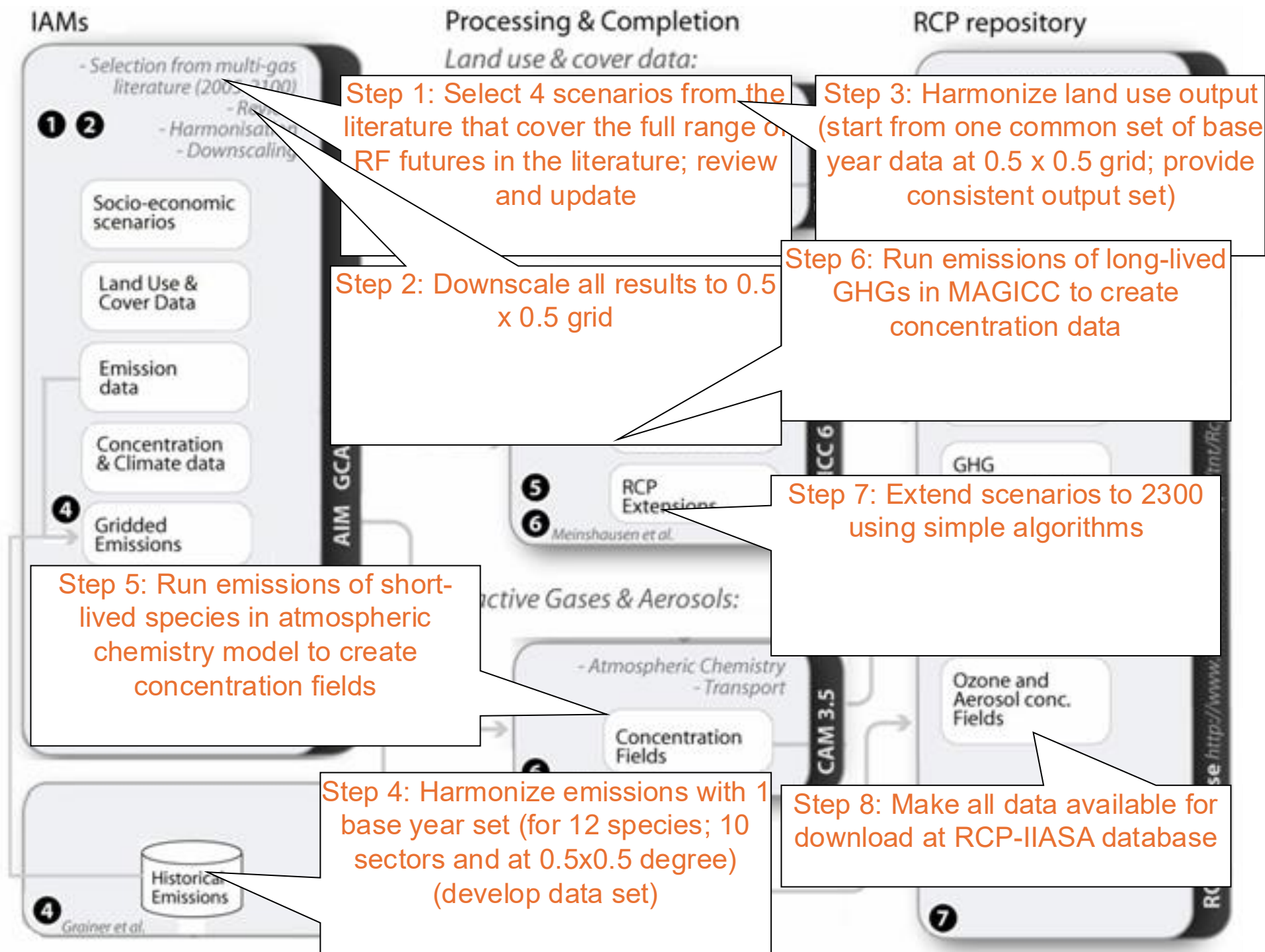
Pathway → the *trajectory* of these concentrations (and their warming effect) through the 21st century.



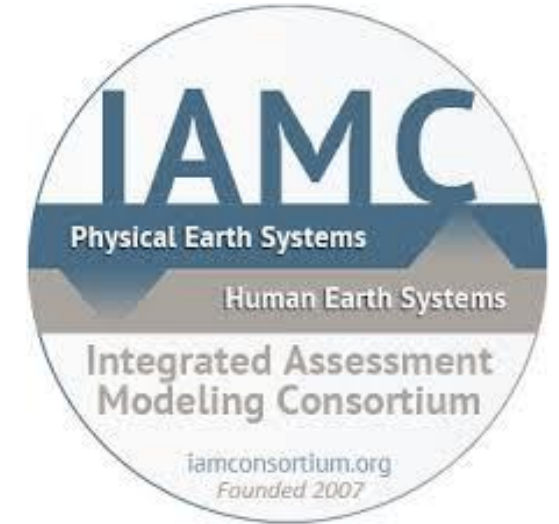
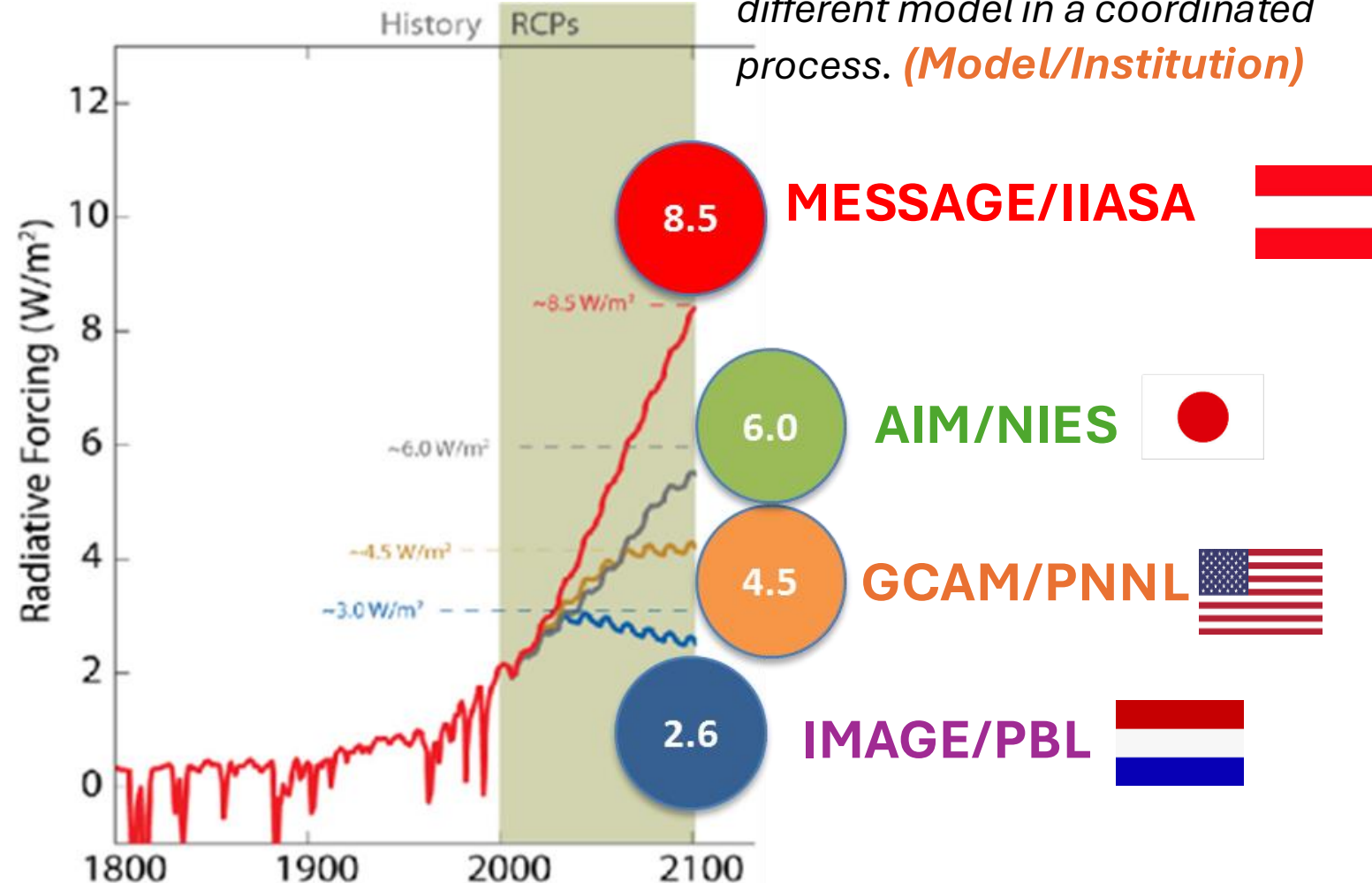
Overall RCP framework of experiments:

*IAMs provide **emission scenarios** → GCMs simulate **climate outcomes** → foundation for RCPs*





Integrated Assessment Models were used to prepare the RCPs:



- Created by the Integrated Assessment Modeling Consortium (IAMC)
- For the intergovernmental Panel on climate change

Some handy equivalents

- CO₂ is measured in both tons of CO₂ and tons of C
- 3.667 ton CO₂ = 1 tons C
- \$1/tCO₂ = \$3.67/tC

CO2-equivalent (ppm)	Radiative Forcing (W/m ²)	Avg. Global Temp. Change Long-term (ΔT)*
1360	8.5 (RCP)	6.8 °C
1030	7.0	5.6 °C
850	6.0 (RCP)	4.8 °C
650	4.5 (RCP)	3.6 °C
550	3.7	2.9 °C
450	2.6 (RCP)	2.1 °C
400	1.9	1.5 °C

* Assumes a climate sensitivity of 3°C. Climate sensitivity is the number of degrees the planet would warm in the long term if the concentration of CO₂ doubled.

RCP8.5 (Riahi et al. 2011)

Core Assumptions

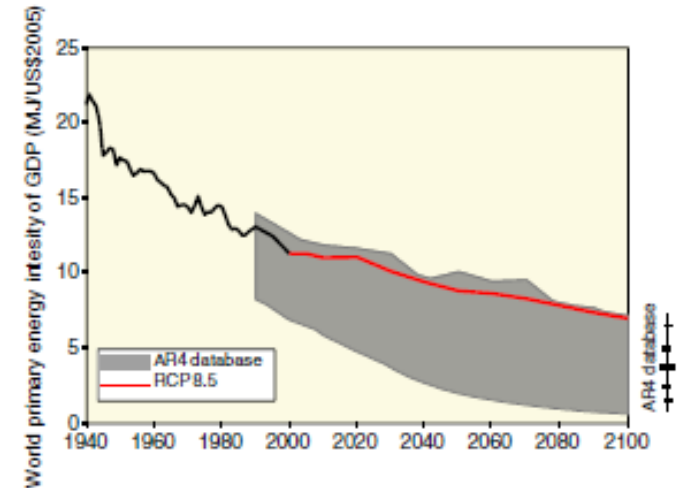
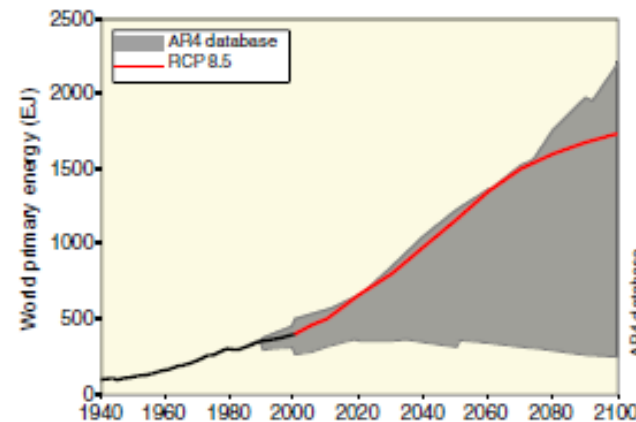
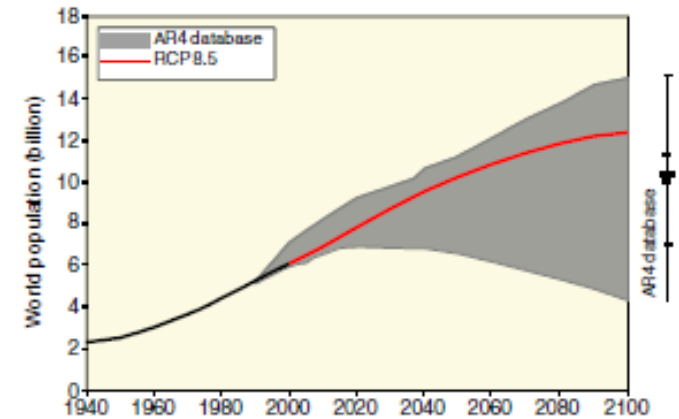
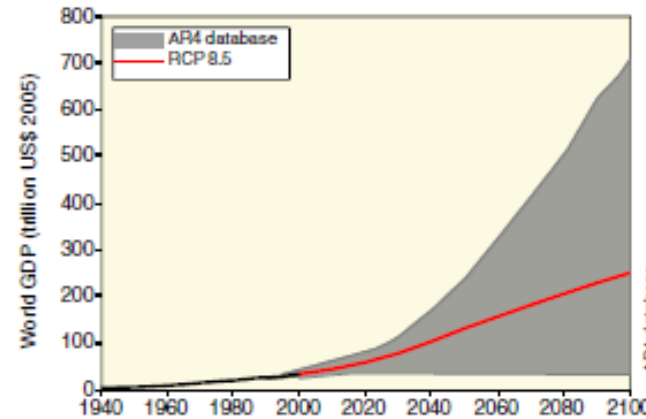
- 🌍 Population: ~12 billion by 2100
- 📈 Slow economic growth, weak globalization
- ⚡ Low tech progress, poor efficiency gains
- ❓ Fossil-fuel dominated, coal expands
- 🚫 No climate policy

Outcomes by 2100

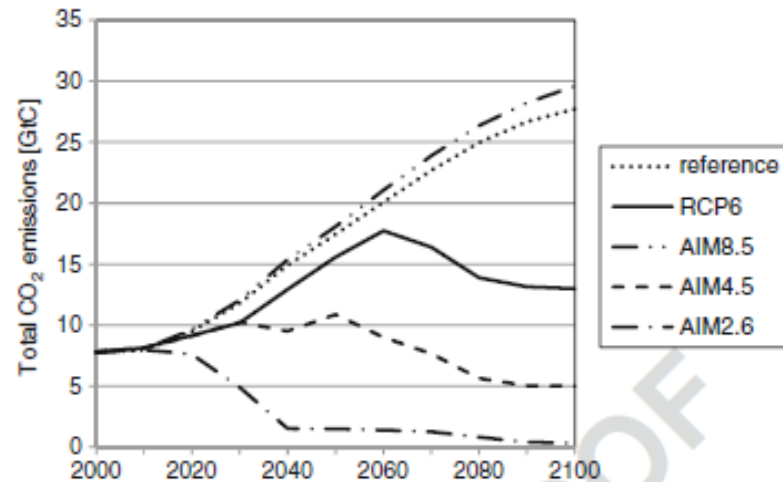
- 🌊 >90 GtCO₂eq/year emissions
- ☀️ +8.5 W/m² radiative forcing
- 🌡️ ~936 ppm CO₂eq concentration
- 🔥 >4°C warming (multi-model avg.)

Why It Matters

- High-end **baseline** scenario
- Tests the **worst-case fossil-intensive future**



RCP6.0 (Masui et al. 2011, AIM) --> a **stabilization pathway**



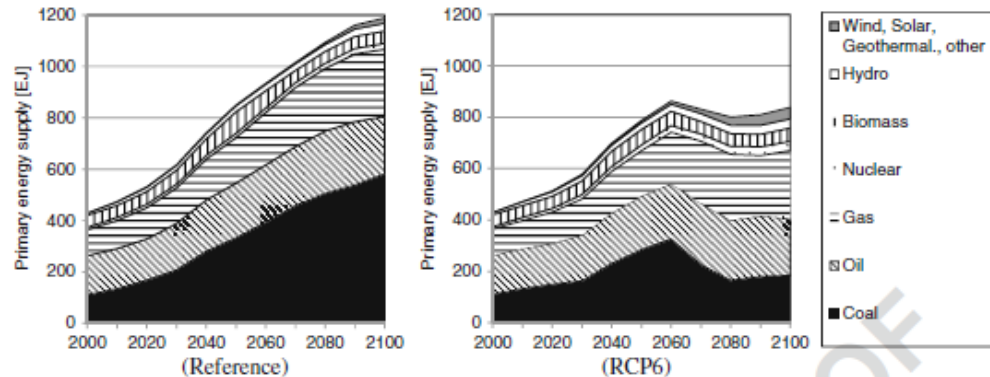
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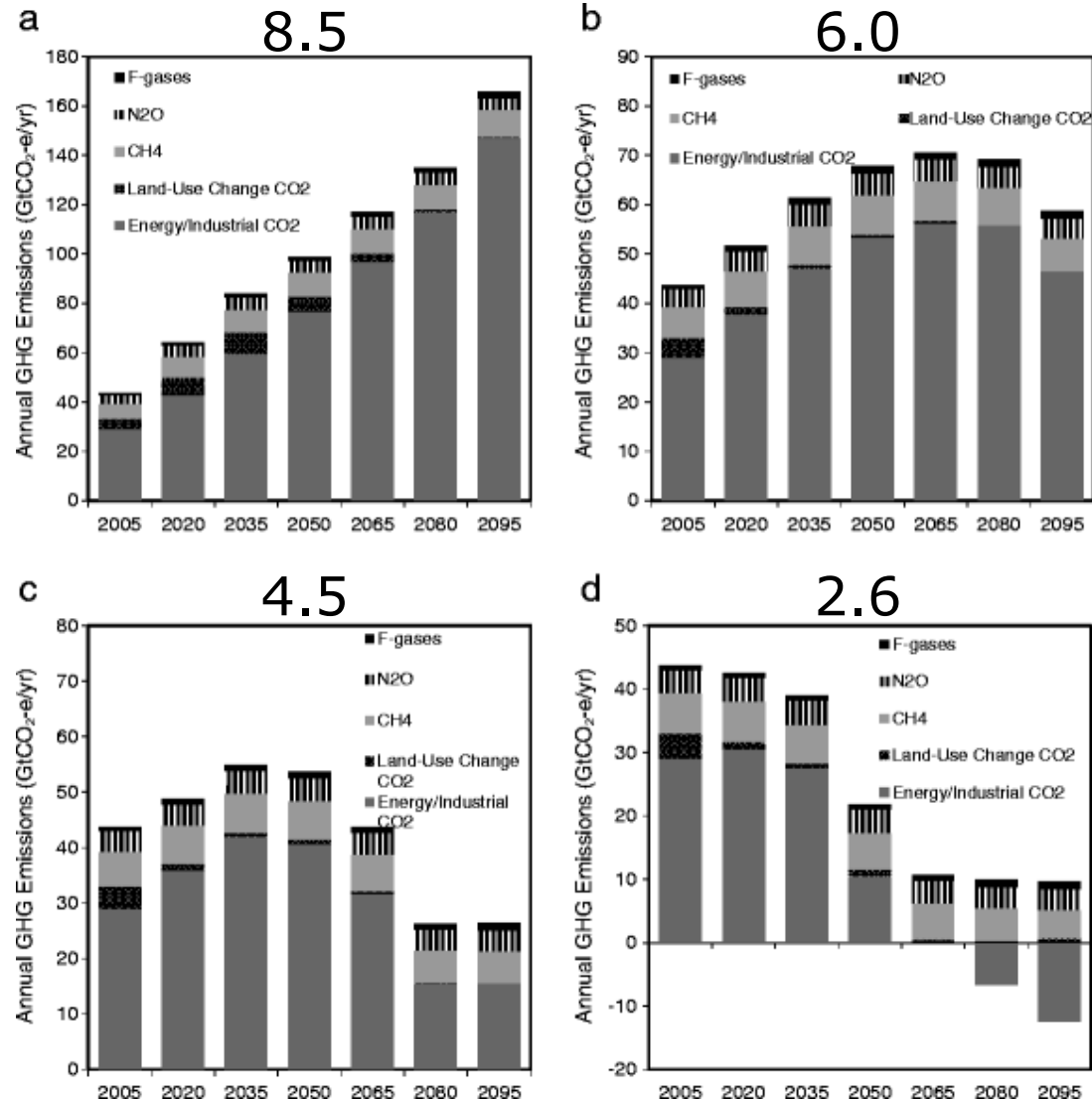
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Shows the consequences of a modest climate target (reductions in 2nd half century; mostly by reducing coal use) [in terms of radiative forcing, concentrations – this scenario can also be interpreted as a medium/low baseline]

RCP4.5 (Thomson, GCAM) --> *Stabilization pathway with moderate policies, ~2–3°C warming.*



Core Assumptions

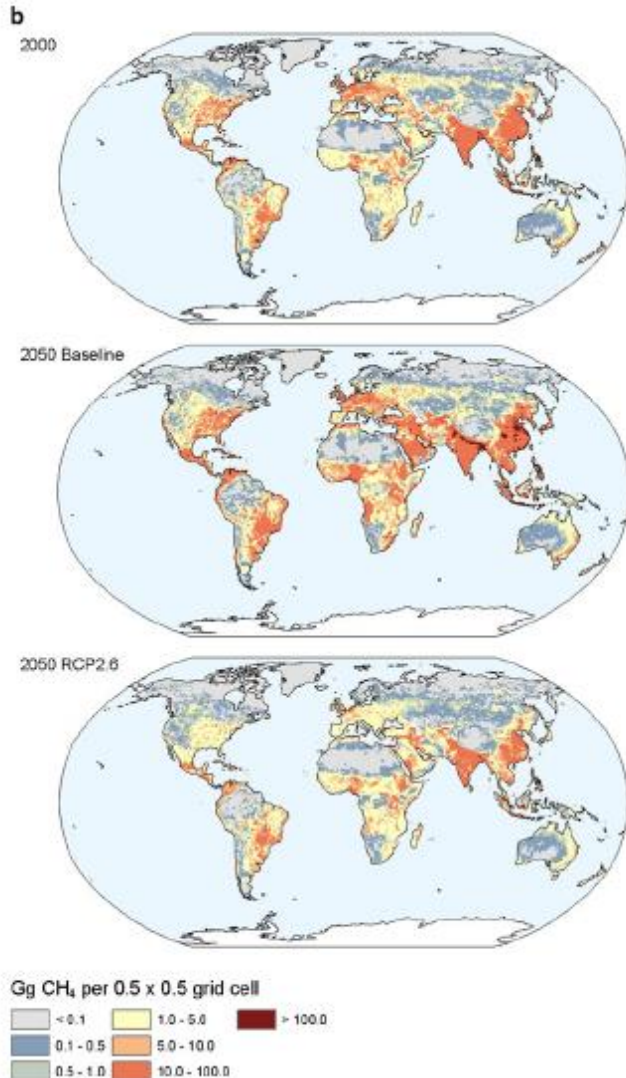
- 🌍 Population: ~9.5 billion by 2100
- 📈 Steady economic growth, moderate globalization
- ⚡ Strong energy efficiency, renewables expand steadily
- 🔋 Balanced energy mix: fossil, nuclear, renewables
- 📄 Moderate climate policies assumed

Outcomes by 2100

- ☀️ Radiative forcing: ~4.5 W/m²
- 🌊 Concentration: ~650 ppm CO₂eq
- 📈 Emissions: peak around 2040, decline afterwards
- 🌡️ ~2.4°C warming (multi-model avg.)

Shows the consequences of medium stringent climate policy – and particularly the possible role of the terrestrial carbon for mitigation

RCP2.6 (van Vuuren, IMAGE) -> *Ambitious climate action, net zero by 2050, ~2°C pathway.*



Core Assumptions

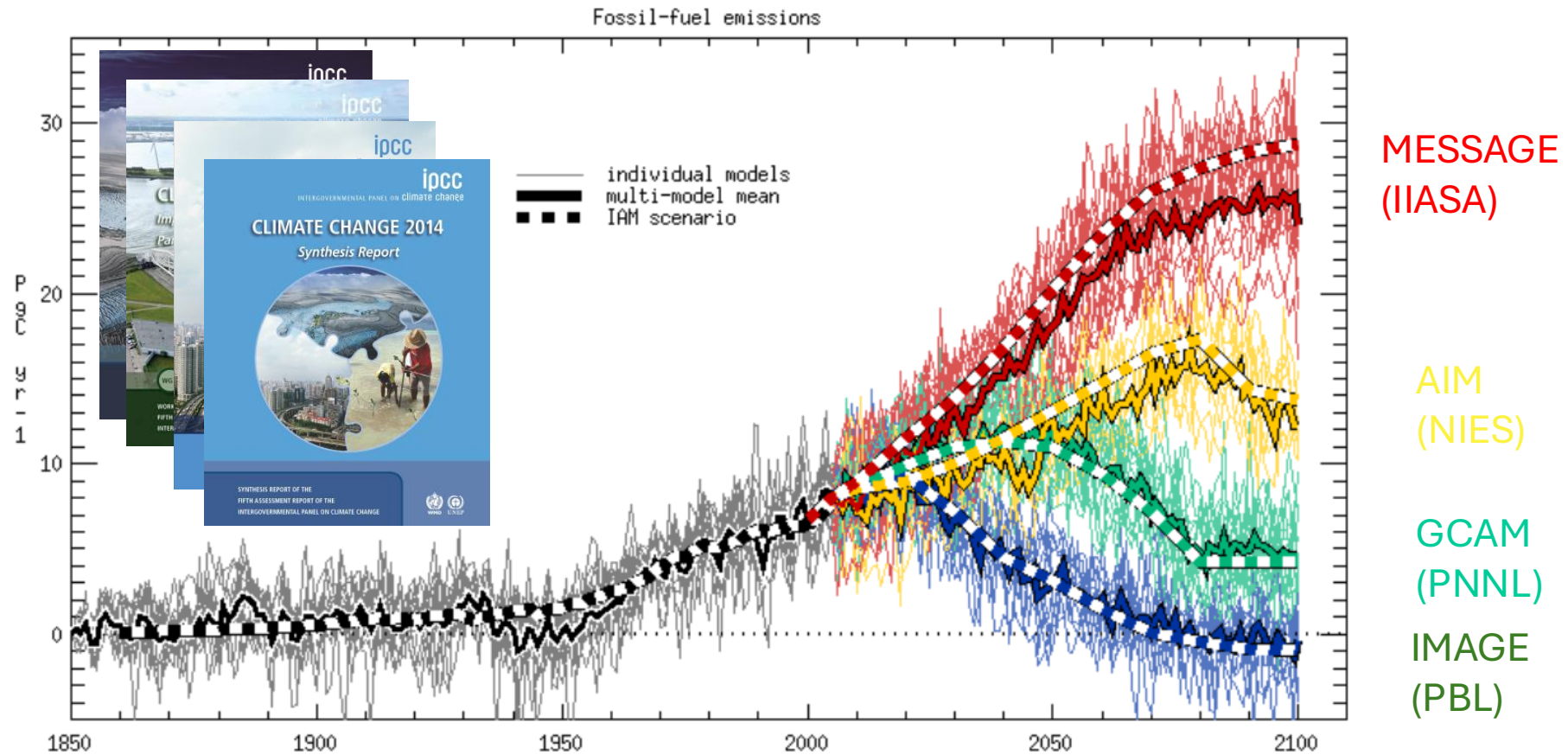
- 🌍 Population: peaks mid-century, ~9 billion by 2100
- 📈 Strong economic growth, rapid tech progress
- ⚡ Major efficiency gains, renewables + nuclear scale up fast
- 📄 Aggressive climate policy to cut GHG emissions deeply

Outcomes by 2100

- ☀ Radiative forcing: peaks ~3 W/m², declines to 2.6 W/m²
- 🌊 Concentration: ~421 ppm CO₂eq
- 📉 Emissions: peak 2020–2030, then steep decline → net negative by late century
- 🌡 ~1.5°C–2°C warming (multi-model avg.)

Only RCP consistent with **Paris 2°C goal**; requires global cooperation.

RCPs were run by climate models and assessed in AR5



In-Class Activity for Database Exploration

- RCP Database: <https://tntcat.iiasa.ac.at/RcpDb/>
- Miro Board:
https://miro.com/welcomeonboard/Yms1NVZ1dktCMDJOQUhXLzJFc0lMYjlCeGpBa1NuWVR1eGQ0STYzS3djdmUW16WGpsUUdDYjhHT0ptMmJrai9wdjFnMnd4WWNFZm9XaW1QQkl0MmhuanUdjlDdFMyR1hoeGF1N0JhMTR5WXZvdVBMNHRwR3g3Q3BTejZRS21QdGo1ZEV3bUdPQWRZUHQzSGl6V2NBPT0hdjE=?share_link_id=201235877272